A Clock-Independent, Time-Domain Rapid Calibration Method for Memristor-based Analog Computing AI Processors

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***Abstract*—Memristor-based analog computing shows promising energy efficiency compared to digital counterparts. High- performance memristor-based AI processor has multi-level memristors, combined with the effect of device retention and ageing, the memristor readout system has tight readout margin to accurately determining the state of the memristor. Because the accuracy of the readout comparator determines the maximum accuracy the system can achieve. To address this, we proposed a high precision, clock-independent offset calibration scheme. The calibration scheme uses edge-triggered pulse-based calibration signal that makes the calibration accuracy independent of clock speed. The coarse-fine calibration steps are used to rapidly converge to minimum offset with high precision. A built-in digital calibration flag indicates the calibration status, that enables the synchronisation across arrays of comparators to facilitate scaling. The comparator is designed, laid out and simulated using 12 nm FinFET process. The comparator achieves average input voltage offset of 120 μV with 50 μV standard deviation after calibration, while the conventional method has the offset of 1.35 mV with standard deviation of 390 μV.**

Index Terms — comparator calibration, analog computing, memristor readout

I. Introduction

Recent advances in in-memory and analog computing are enabling ultra-low-power Edge-AI processing [1] [2] [3]. This is particularly compelling for power-constrained platforms: while many digital edge processors operate at watt-level power [4], biomedical ICs often target milliwatt budgets [5] [6], making on-device computation difficult. By leveraging charge-domain and device-physics-level operations, in-memory/analog approaches can reduce energy by orders of magnitude, supporting intelligent sensing and inference in battery-powered biomedical nodes with minimal overhead [7] [8].

Among candidate technologies, memristor-based computing is widely studied due to its non-volatile, multi-level programmability and high integration density. Memristors can be electrically programmed to set persistent conductance states, enabling co-located storage and computation in crossbar arrays. In such arrays, matrix–vector multiplication is performed in situ by applying input voltages/currents to programmed conductances and sensing the resulting summed currents/voltages along orthogonal lines, providing an efficient primitive for neural inference.

For binary and multi-level operation, the readout comparator is pivotal in resolving the memristor state (Fig. 1(a)). In memristor-based AI processors—where storage and computation occur within the array—read accuracy governs both training observability and reliable retrieval of programmed weights. Comparator input offset directly translates into state-decision errors, and the stringent read margins of multi-level devices make low offset essential. Ageing further compresses the resistance window, reducing margin and tightening the offset requirement; consequently, robust operation demands a comparator with minimal offset across lifetime and operating conditions.

To suppress input-referred offset, comparators typically employ pre-operation calibration, broadly categorized as trimming or dynamic schemes. Trimming reduces offset by permanently adjusting the input pair, commonly via laser or e-fuse programming [9] [10]. Although attractive as a one-time correction, trimming is irreversible and cannot track environmental drift (e.g., temperature). This limitation is acute in memristor AI accelerators, where activity-induced self-heating can shift device parameters and exacerbate residual offset, ultimately degrading readout fidelity and system-level inference accuracy.

In contrast, dynamic calibration provides adaptive offset compensation by enabling real-time re-calibration during operation. Similar to auto-zeroing, dynamic calibration is done by sampling the offset voltage in a capacitor, then subtracting that voltage to the input or the output of the comparator to counter the offset. This work adopts the input-based dynamic calibration method, as the output-based method alters the impedance, introducing parasitic loading and reducing the resolving speed. These effects are undesirable for a high-speed, high-performance readout system.

A key design consideration of the calibration system in a memristor readout array is that the calibration speed has to be rapid. Since periodic recalibration is required to track the offset drift caused by factors such as temperature, the calibration must be fast enough to not interfere with the overall performance of the system.

The circuit of a typical dynamic calibration comparator is shown in Fig. 1(b). with the timing diagram shown in Fig. 2(a). The output of the comparator is used to control the charge pump, which charges and discharges the calibration capacitor C_{CAL} that is connected to one side of the comparator inputs [11]. The voltage stored in C_{CAL} is then used to cancel the voltage offset inside the comparator input pair.

The structure of the paper is as follows: Section II describes the system overview, Section III shows the control circuit in detail, Section IV shows the simulation results and comparison between the proposed calibration scheme and the conventional dynamic calibration scheme. Conclusion and discussions are in Section V.

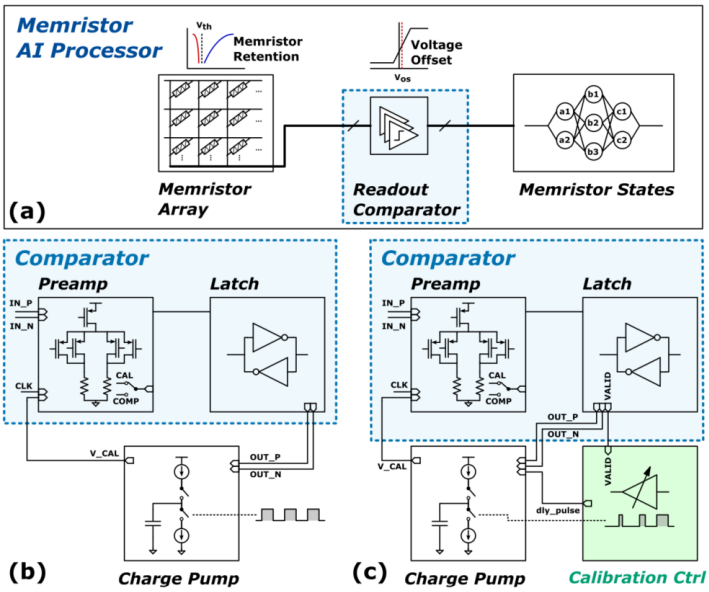


Fig. 1: Block diagram of (a) Memristor AI system and summary of performance limiting factors; (b) comparator with conventional dynamic calibration scheme; (c) comparator with proposed dynamic calibration scheme

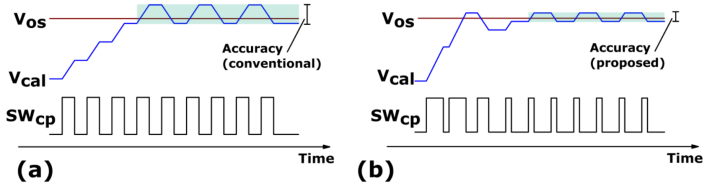


Fig. 2: Timing diagram (a) comparator with conventional dynamic calibration scheme; (b) comparator with proposed dynamic calibration scheme

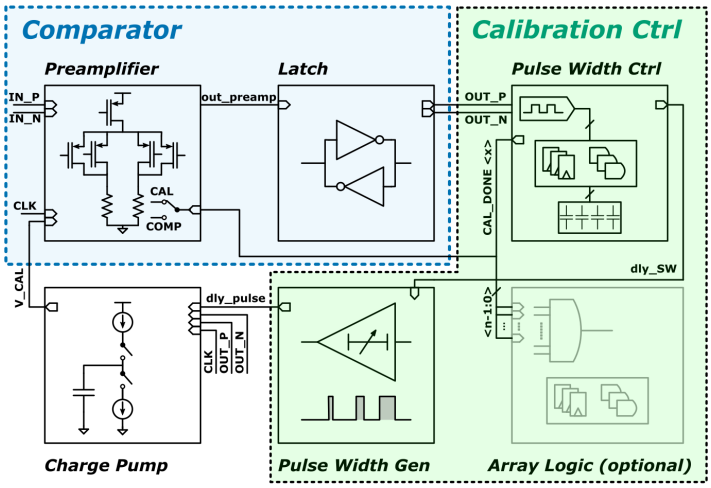


Fig. 3: Block diagram of the proposed system. The comparator output is connected to the pulse width controller. The pulse width controller determines the pulse for the charge pump to achieve coarse-fine calibration

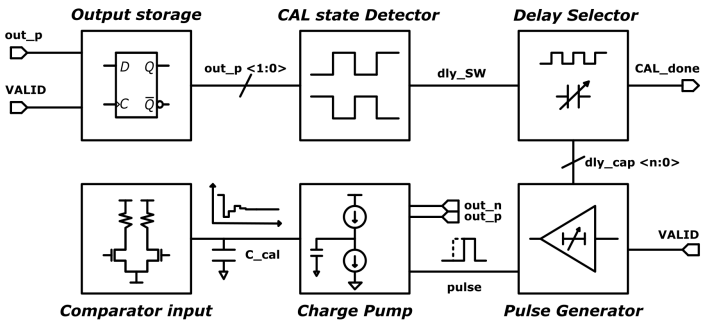


Fig. 4: Block diagram of comparator calibration circuit

II. System Overview

Figure 1(c) illustrates the system-level architecture of the proposed calibration scheme. This method employs a coarse-fine pulse-width modulated (PWM) signal to drive the charge pump for offset calibration. Compared to driving the charge pump directly with the comparator output, using fixed-width pulses has several advantages.

First, the rising-edge-triggered pulse is independent of the clock frequency, this means that the calibration accuracy remains consistent regardless of clock frequency. Second, the pulse width can be easily tuned, enabling adjustable calibration resolution. The timing diagram in Fig. 2(b) demonstrates how calibration accuracy improves by modulating the pulse width.

The detailed comparator calibration flow is depicted in Fig. 3. The comparator latch output feeds into a pulse width control unit that monitors the calibration progress and determines whether the target accuracy has been achieved. Based on this, the control unit dynamically adjusts the pulse width by selecting delay-control capacitors with progressively smaller capacitance values. This pulse width controls a pulse generator that drives the charge pump, which in turn performs the input offset calibration.

Since the pulse width is governed by the pulse generator rather than the input clock (*CLK*), calibration accuracy is immune to variations in clock frequency. The pulse width control unit also generates a *CAL_DONE* flag, asserted when the minimum offset level is reached at the narrowest pulse interval. This flag is particularly useful for large comparator arrays, where synchronization of calibration status across multiple units is critical—a topic discussed in more detail later.

III. Control Circuit Design

Figure 4 presents the block diagram of the calibration control system. Initially, the calibration state (*CAL*) is monitored by recording the current and previous comparator outputs (*out_p*). These outputs are then processed by a minimum accuracy detector, which verifies whether the calibration has achieved the resolution target for the current configuration. Based on this assessment, the delay selector adjusts the delay accordingly, thereby controlling the pulse width used for the calibration process.

The *CAL_DONE* signal is a useful feature for comparators working in a large array. Many memristor-based Analog AI processors have a large number of readout comparators running in parallel, which need to be calibrated at the same time. Therefore, the *CAL_DONE* flag can be connected to an *AND* logic which will be asserted when all comparators in the array have finished their calibration. Also, *CAL_DONE* flag can be used to drive a switch that automatically switches the comparator from calibration mode to comparison mode as soon as calibration is complete.

The variable pulse generator circuit [12] is shown in Fig. 5. The current-starved inverter has a fixed source current controlled by $M3$, combining this with a varying value of delay capacitor C, a controllable delay can be generated. The delayed pulse then passes through a *NAND* gate with the undelayed signal, the time difference will generate a *HIGH* pulse at the output of the *NAND*

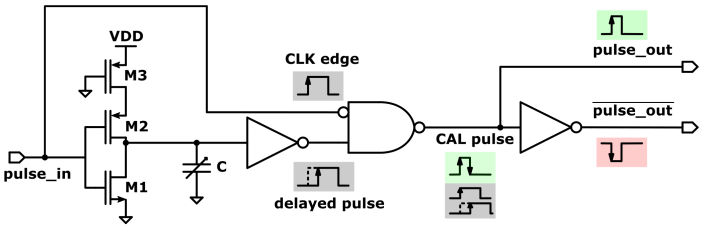


Fig. 5: pulse generator circuit derived from [12], the pulse width is set by the current starve inverter and binary-weighted capacitors

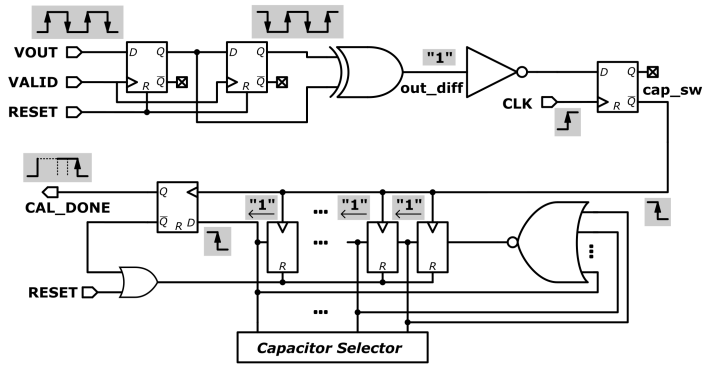


Fig. 6: Delay control logic, this circuit selects the capacitor in the delay generator if continuous signal is detected from the comparator output

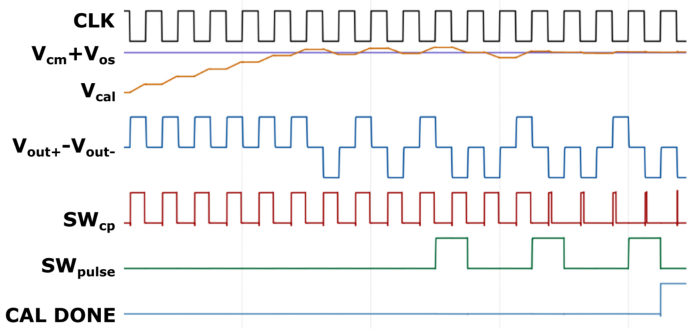


Fig. 7: Timing diagram of the calibration process under 200 MHz clock. When the alternating output pattern starts to appear, it triggers the SW_{pulse} signal, which switches to a shorter calibration pulse to achieve higher calibration accuracy

gate. The differential *pulse_out* signal is generated to gate both the PMOS and the NMOS in the calibration charge pump.

The delay is selected by a series of sequential logic, it takes the comparator output and uses it to determine the pulse length used for the calibration. Fig. 6 shows the delay control logic, it has a comparator output state detector followed by a capacitor selector that determines the delay (clocked by *cap_sw*). The detailed waveform of one propagation is shown on top of the logic blocks.

Fig. 7 shows the timing diagram of the control logic in Fig. 6. The calibration initially starts with large steps and long pulses, after it reaches the condition to trigger the SW_{pulse} logic, a smaller capacitor is connected to the pulse generator to generate a shorter pulse for more accurate calibration. After all pulse width options have been exhausted, which is three pulse widths in this case, the control logic will output a *CAL_DONE* signal to indicate the calibration process is complete.

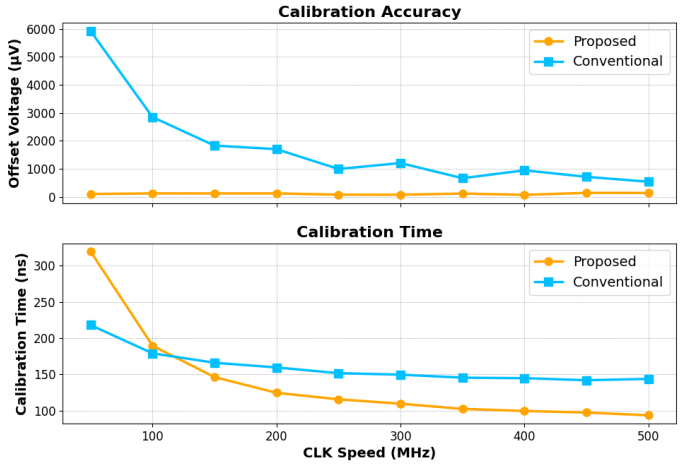


Fig. 8: Calibration time and accuracy performance of the proposed and conventional calibration scheme, the calibration accuracy of the proposed method is independent to clock frequency, while the conventional method has higher accuracy on faster clock

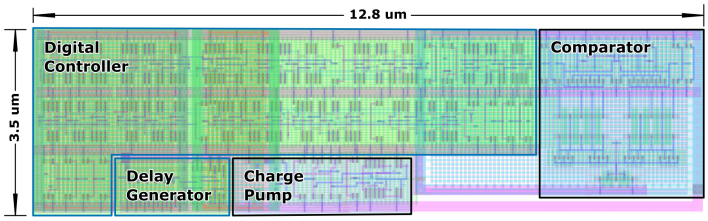


Fig. 9: The layout of the comparator with calibration circuit

IV. Simulation Results

Fig. 8 shows the calibration accuracy (represented by offset voltage) and speed (represented by the time between *RESET* and *CAL_DONE*) across different clock frequencies from 50 MHz to 500 MHz. The proposed method maintains excellent calibration accuracy below 200 μV, while the conventional method failed to achieve good accuracy with slower clocks. The proposed method excels in calibration time when the clock frequency is higher than 100 MHz, however, the speed is not ideal for slower clocks, as the routine must still pass through all the calibration stages.

Fig. 9 shows the layout of the proposed comparator. The comparator with the calibration system has the size of 12.8 μm $\times$ 3.5 μm; the majority of this area is taken by the digital controller. The timing and calibration metal-oxide-metal (MOM) capacitor is on the top layer of the circuits to achieve good area efficiency.

Fig. 10 shows Monte Carlo simulation results of the comparator offset after calibration using both the proposed method and the conventional method. The proposed calibration method has a substantially smaller spread of voltage offset after calibration, with a mean calibration offset of 118 μV. This is 10 times more precise than the conventional method, which has a mean offset of 1.16 mV.

Fig. 11 shows Monte Carlo simulation results of the comparator calibration time using both the proposed method and the conventional method. The proposed calibration method improves the calibration speed by using a smaller value calibration capacitor, as it can use shorter pulse to achieve higher calibration accuracy. The figure shows that the proposed method improves the mean calibration time by 20% with less standard deviation. In an AI

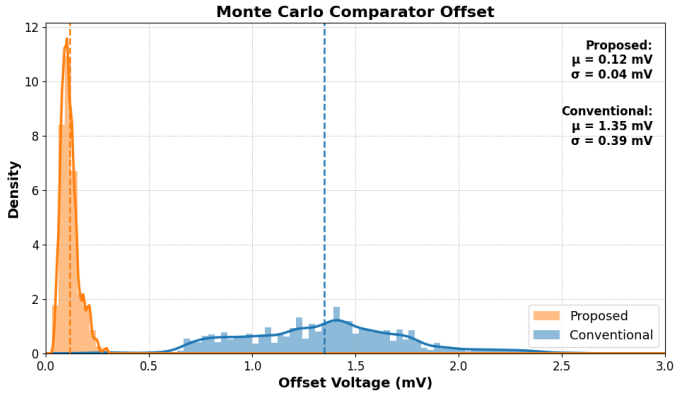


Fig. 10: Monte Carlo simulation of calibration offset. The proposed calibration scheme has the mean offset of 0.12 mV, ten time smaller than the conventional calibration scheme

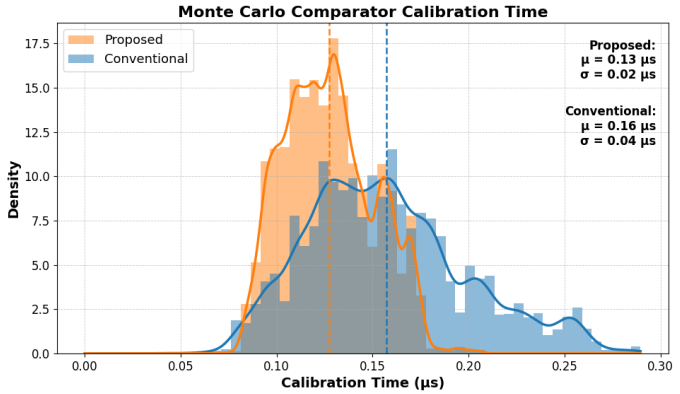


Fig. 11: Monte Carlo simulation of calibration time. The proposed calibration scheme has shorter calibration time with lower standard deviation compared to conventional method

readout system with large array of comparators, the smaller spread is beneficial as all comparators in the array can finish their calibration at similar time, improving predictability of system behaviour.

V. Conclusions and Discussions

This paper presents a novel comparator calibration technique that achieves rapid calibration speed with clock-independent accuracy. The calibration charge pump is driven by a pulse generator with a configurable fixed pulse width, rendering the calibration accuracy insensitive to clock frequency. This fully self-contained dynamic calibration scheme is particularly well suited for memristor readout arrays in AI applications, which demand high-speed operation and low-offset comparators calibrated and operated within large arrays. Simulations demonstrate that the comparator achieves an input-referred offset precision of approximately 120 μV across clock frequencies ranging from 50 MHz to 500 MHz.

Table I compares the performance of the proposed work with prior art implemented in similar CMOS processes. The final column references the pioneering dynamic calibration method introduced in [11]. Our approach improves offset accuracy by an order of magnitude using a comparable calibration technique, achieving an input-referred offset of 120 μV, which surpasses many state-of-the-art designs. Furthermore, the proposed comparator exhibits significantly lower power consumption, enhancing the power efficiency of energy-constrained analog computing systems.

TABLE I: Comparison with other state-of-the-art comparators with calibration. The proposed work excels in its offset accuracy and power consumption

	This work	[13]	[14]	[15]	[11]
Process [nm]	**12**	12	28	14	90
Frequency [Hz]	**50-500 M**	1.5 G	4 G	1 G	250 M
Offset [mV]	**0.12**	1.9	0.14	0.26	1.3
Power [μW]	**10**	240	220	213	40

Future work includes finding the optimal calibration ratio with optimal delay capacitor size in both design and layout, optimising the calibration state detection logic to further reduce the calibration time, and fabricating the design to observe the offset performance in silicon

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